

G1 Campaign 2 — Session Protocol (24 Jul 2026)

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0. Sync (GPUEDGE, 2 min)

```
cd ~/Documents/G1_UNITREE_ROBOT_META_REASONING
git pull --rebase origin main
```

1. Launch the stack (3 terminals)

```
# T1 – perception server (cy=90 for the 16:9 frames; --debug enables the live viewer):
G1_FL00RCOLOR=1 python perception_server.py --host 0.0.0.0 --port 8008 \
--fx 300 --fy 300 --cx 160 --cy 90 --cam-h 1.10 --cam-pitch -10 --debug
# viewer: http://192.168.51.80:8008/view      health: curl -s localhost:8008/health

# T2 – WebView proxy (after the robot ritual below):
ios_webkit_debug_proxy

# T3 – spill marker v2 (leave running ALL session):
python spill_mark.py
```

Marker command	Meaning
ENTER	one spill (a slosh burst = ONE mark)
w 200	record cup weight in grams — BEFORE and AFTER every leg
i reason	mark the current run INVALID (e.g. i marker died)

It heartbeats every 2 s — the run dataset records whether this terminal was alive (gt_hb_seen / gt_alive_at_end / gt_dropouts) and the run console alarms if it dies.

Robot ritual: boot robot → iPhone Bluetooth ON → connect FROM THE APP (Wi-Fi key rotates every boot; never join from iOS Settings) → verify in-app teleop → USB → T2 lists the WebView. Optional injection check: python3 g1_inject_teleop.py sniff then drive (W/S/A/D, SPACE=stop, X=exit).

```
python3 g1_goto.py noisecheck      # 20 s standing still: the day's sensor-noise floor
python3 g1_goto.py reloccheck      # after walking the robot near a waypoint
```

2. Furniture census (REQUIRED before water, ~10 min)

```
# Robot standing still, 1–1.5 m FACING the sofa arm, NO people in the camera view:
G1_PERC=127.0.0.1:8008 python3 g1_goto.py furniture 15
# Lists candidate cells and asks confirmation (s/N) — check they fall ON the sofa.
# Repeat from a 2nd angle; then the same facing the low cabinet.
# Mistake? python3 g1_goto.py furniture undo
git add nav_map.json && git commit -m "furniture census" && git push
```

3. DRY SHAKEDOWN — one pair WITHOUT water, ALWAYS before pouring

Yes — no water until this passes. 16 fixes landed since the last dry pass. One A↔B pair with an empty cup verifies door crossing, the new door-zone guard, the camera brake and the instrumentation, at zero cost. Only then pour.

```
G1_ARM=SHAKEDOWN G1_META2=2 G1_M2_CFG=config_meta2_g1payload.json G1_M2_REALCAL=1 \
G1_M2_STATE=meta2_state_c2b_shake.json G1_M2_STATE_INIT=meta2_state_twin_explored.json \
G1_M2_L3=1 G1_M2_L4=1 G1_M2_PROFILES=1 G1_M2_D0ORLIB=1 G1_M2_FRAGSPEED=1 \
G1_PERC=127.0.0.1:8008 python3 g1_goto.py gotoviz B      # then: gotoviz A
```

PASS checklist: reaches both ways · no door contact (door_center event with gap≈0.85) · door_guard events near the vano (that is the new deterministic floor working, not a fault) · perc_n in the dozens · no unexpected stops in open space · marker heartbeat alive. Push + report in chat — 2-minute log check before water.

4. Data runs — pairs at 200 g, ABBA order

Weigh with the marker (w 200 before leaving, w <result> on arrival). **Never refill between the two legs of a pair.** Pair order (ABBA kills environment drift): **GOV, BASE, BASE, GOV, GOV, BASE.** A governed HELP stop is a VALID result: weigh, note, continue. Marker dies mid-run → i marker died, relaunch marker, redo that run.

```
# GOVERNED pair (C2-GOV) — keep the same state file all day:
G1_ARM=C2-GOV G1_FILL_G=200 G1_CUP_G=246 \
```

```
G1_META2=2 G1_M2_CFG=config_meta2_g1payload.json G1_M2_REALCAL=1 \
G1_M2_STATE=meta2_state_c2b_m2.json G1_M2_STATE_INIT=meta2_state_twin_explored.json \
G1_M2_L3=1 G1_M2_L4=1 G1_M2_PROFILES=1 G1_M2_D00RLIB=1 G1_M2_FRAGSPEED=1 \
G1_PERC=127.0.0.1:8008 python3 g1_goto.py gotoviz B          # leg 2: gotoviz A

# BASELINE pair (C2-BASE):
G1_ARM=C2-BASE G1_FILL_G=200 G1_CUP_G=246 \
G1_META2=0 G1_PERC=127.0.0.1:8008 python3 g1_goto.py gotoviz B      # leg 2: gotoviz A
```

If time allows — the headline pair: repeat both commands with `G1_FILL_G=247` (full cup: the condition that aborted 4/4 in campaign 1; the governed turn cap should now survive it).

5. Photo & video catalog (phone; for the paper/viva)

Folder `docs/photos_c2/`, filenames `YYYYMMDD_<s>lug>.jpg`:

#	Shot
1	Robot front + side WITH the cup mounted (payload placement)
2	Cup on the scale: one "before" (200 g) and one "after" reading
3	The door/vano from the A side and from the B side
4	The sofa corridor (laser-blind obstacle) and the low cabinet
5	Waypoint A pocket and goal pocket B
6	Operator station: marker terminal + /view page on screen
7	Action: robot mid-door-crossing; robot arriving at B
8	Wide shot of the full route

```
# What the robot sees (annotated depth+Y0L0 frame – run while the robot faces the sofa):
curl -s localhost:8008/debug.jpg -o docs/photos_c2/$(date +%Y%m%d)_perception_view.jpg
```

Video (30–60 s clips): a governed start with water (turn governance visible), one full door crossing, and — if it happens — a governed HELP stop.

6. After every block

```
git pull --rebase origin main
git add dataset/ runs_summary.csv meta2_state_c2b_*.json goto.log nav_map.json docs/photos_c2
git commit -m "C2-GOV pair 1: 200g, weights 200->161->148"
git push
```

...and report weights + marks per leg in chat for the live accounting.

Priority if time runs short: census → dry shakedown → GOV+BASE pairs (min 2+2) → photos 1-3-7 → FULLCUP 247 g.